

Over-sampled Data Converters – ET4278

Practical Session

May 24th, 2019

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Today Overview

- Solutions to Assignment
- Practical
 - Continuous-Time (CT) modulators
 - Multi-bit modulators: DEM
 - Cascaded modulators (MASH)
 - Bandpass modulators

Today Overview

- Solutions to Assignment
- Practical
 - **Continuous-Time (CT) modulators**
 - Cascaded modulators (MASH)
 - Bandpass modulators
 - Multi-bit modulators: DEM

Exercise 2.7

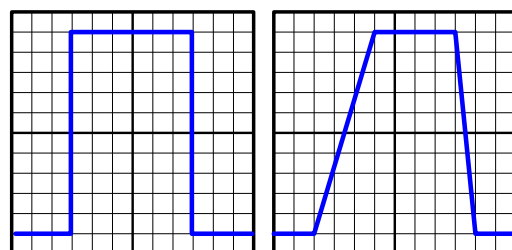
- Download (from Blackboard) and run the m-file: MOD2_CT.m
 - This file models a 2nd order CT sigma delta modulator with a feed-back topology.
 - Locate the two “for” loops simulating the modulator and the internal steps of the continuous time loop filter.
 - Locate the integrator unity gain frequency definitions, and try to trace their relation to the calculation step times.
- Modify the code to implement a feed-forward modulator
 - Size the modulator to reach specs of Assignment 3 (SQNR = 100 dB, SNR = 90 dB in BW=20 kHz)
 - Scale the coefficient to keep the integrator outputs < 0.5V

Exercise 2.7 - Solution

- Use the same OSR as discrete-time implementation and scale coefficient to get same shaping
 - $OSR = 250; f_b = 20; \Rightarrow f_s = 10 \text{ MHz}$
 - $\omega_{a1} = 2\pi f_s / (f_s \cdot nr_steps)$
 - $\omega_{a1} = 2\pi f_s / (f_s \cdot nr_steps)$
 - $c1 = 4\pi$
- Add noise:
 - $noise_rms = amplitude \cdot \sqrt{f_s \cdot nr_steps / 4 / f_b / 10^{(90/10)}}$
- Scale:
 - $b1 = 0.05;$
 - $b2 = 0.055;$

Waveform asymmetry of DAC

- A practical single-bit DAC will in most cases output signals with asymmetric rise and fall times.
- This leads to a problem called: Intersymbol Interference (ISI) in continuous time sigma-delta modulators.
- ISI results in distortion, and more noise in the base band due to the intermodulation of harmonics with shaped Q-noise.

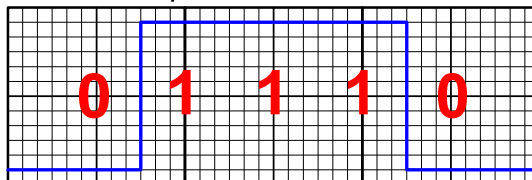


Ideal DAC signal

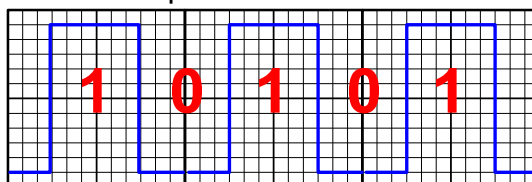
Asymmetrical DAC signal

No ISI

"01110" sequence



"10101" sequence

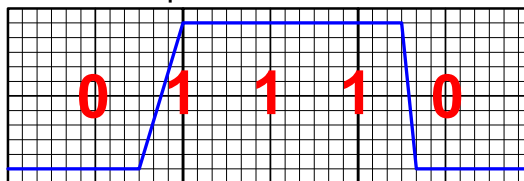


Ideal DAC signal

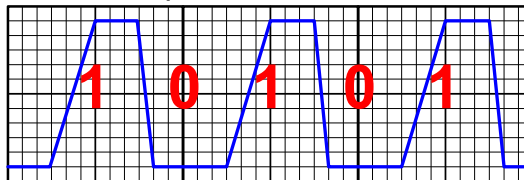
- Ideal DAC $\Rightarrow$ energy (area under the pulses) is the same for both "01110" and "10101" sequences.

ISI

"01110" sequence



"10101" sequence



Asymmetric DAC signal

- Non-ideal DAC $\Rightarrow$ energy of "01110" and "10101" sequences is **NOT** the same $\Rightarrow$ **data dependent** errors!

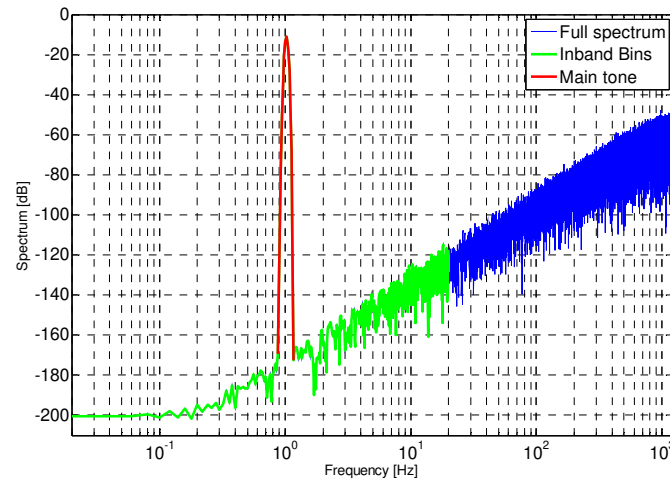
Exercise 2.8

- Download (from Blackboard) and run the m-file: MOD2_CT_NRZDAC.m
 - This m-file simulates a 2nd order continuous time **feed-forward** modulator with a sinusoidal input signal.
 - The modulator's single-bit DAC has **asymmetric** rise and fall times.
 - Go through the m-file, and observe how the CT DAC signal is defined and how the asymmetric rise and fall times are generated.
 - Run the code with asymmetrical DAC signals (default), and with ideal DAC signals (by removing the comments at proper place). Observe the output spectrum. What happens? Why?
 - Calculate the SNR and see how much resolution you lose due to the DAC non-ideality.

Today Overview

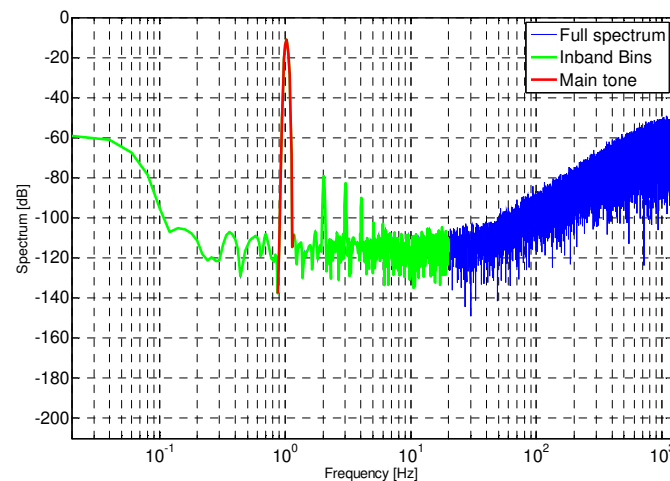
- Solutions to Assignment
- Practical
 - Continuous-Time (CT) modulators
 - **Multi-bit modulators: DEM & DWA**
 - Cascaded modulators (MASH)
 - Bandpass modulators

Multi-bit modulators



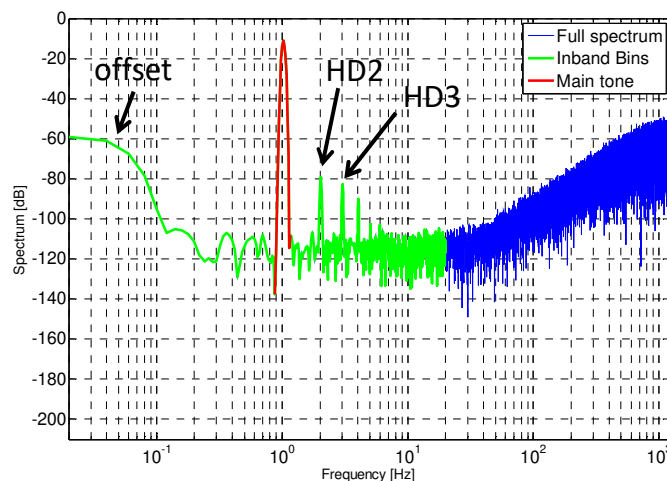
- Elements = [1 1 1 1 1 1] $\Rightarrow$ SNR = 91dB

Multi-bit modulators



- 0.5% **random** mismatch $\Rightarrow$ SNR = 49 dB (in this case)

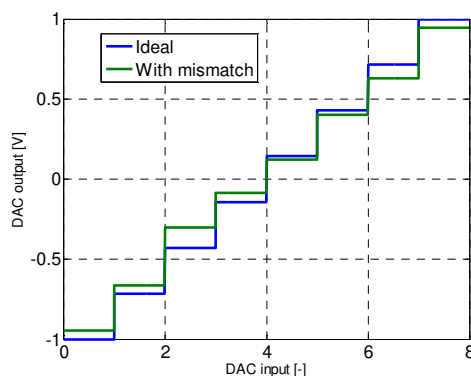
Multi-bit modulators



- It is better to define SNDR (Signal-to-Noise-and-Distortion-Ratio)

DAC Mismatch

- The mismatched applied to DAC elements translates into a non-linear DAC characteristic:
- DAC non-linearity is not suppressed by the loop gain, since the DAC output is directly subtracted from the input signal.

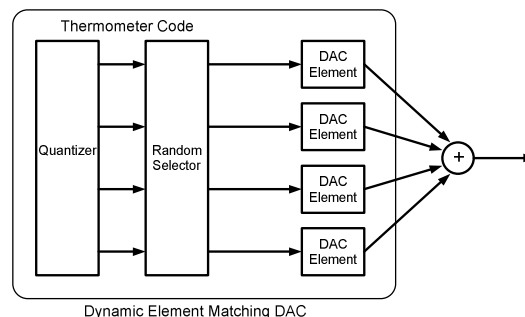


Dynamic Element Matching (DEM)

- Mismatch in a thermometer DAC leads to a non-linear characteristic.
- This non-linearity results in tones at the output of the modulator.
- If the same set of unity elements are always used to generate a certain input code, the mismatch error at the DAC output will be correlated with the DAC input.
- To break this correlation, we can use DEM.
- DEM ensures that all elements are used equally often, in a relatively random manner, and as such the DAC's characteristic is averaged.
- This means that DAC mismatch is turned into a noise-like signal, which in some cases can even be 1st or 2nd order noise-shaped!

Randomization

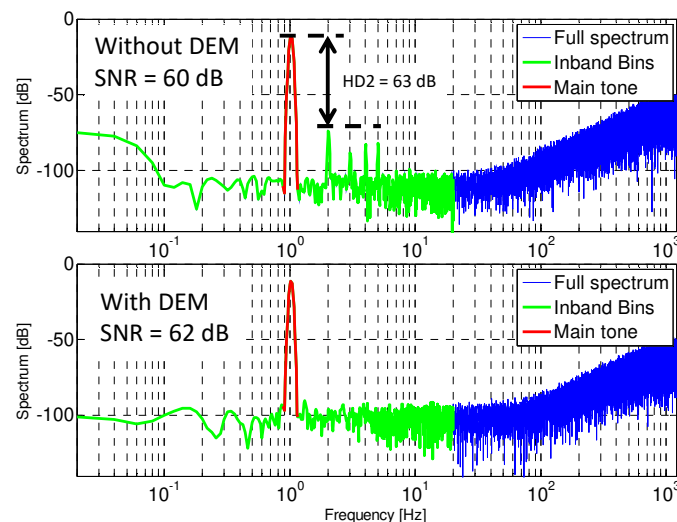
- This involves picking a random group of unit elements to generate the DAC output
- A random selector is placed between the quantizer and the DAC.



Exercise 3.1

- Run the m-file: MOD2_3Bit_ThermoDAC.m
 - This code implements a 2nd order modulator with thermometer code DAC made of mismatched unit elements.
 - Change the code to use the function thermoDACrnd.m (uncomment a line). Check the code thermoDACrnd.m, used by this code and figure out how the unit elements are randomized per sample.
 - The modulator applies randomization to the DAC elements.
 - This means that per sampling period of the modulator every time a different group of unit elements are used in order to make the DAC output signal.
- Run the code and compare the output FFT and SNR of this modulator to the case of the modulator in previous exercise when the DAC was mismatched and no dynamic element matching technique was used.
- How much does it improve the SNR? What happens to the tones? Why?

Exercise 3.1 - Discussion



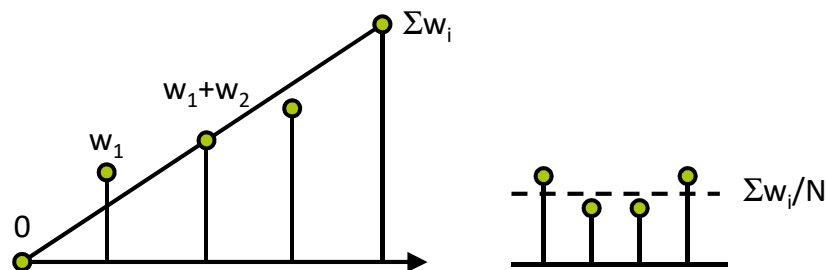
Data Weighted Averaging (DWA)

- The DAC elements are selected sequentially from an array.
- A pointer points to the index of the next unused DAC element in the array.
- The position of the pointer is updated depending on the input code.

Example of an 8-level DAC incorporating DWA

Cycle	DAC	DAC elements							
1	2								
2	2								
3	5								
4	2								
5	3								
6	4								

How DWA works (1)



- Consider a single-ended DAC with N elements: $w_1, w_2, \dots, w_N$
- Ideally, $w_i=1$
- DWA cannot change the endpoints at 0 and Σw_i
- But the weights of a real DAC $w[n]$ are evenly distributed around the average weight $\Sigma w_i/N \Rightarrow \Sigma(e[n]) = 0$.

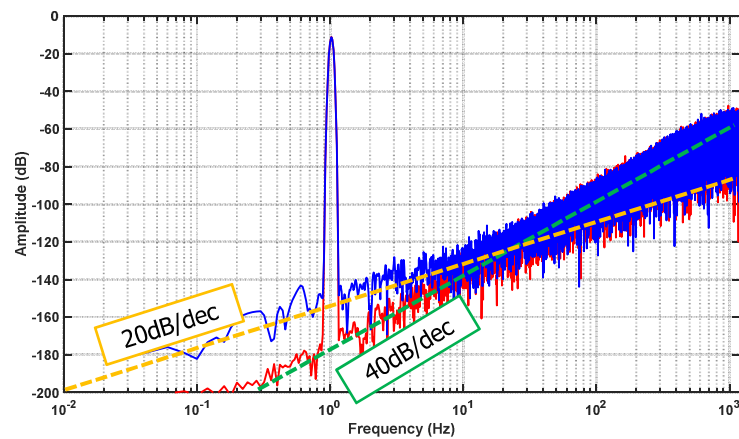
How DWA works (2)

- Error of the DAC at cycle n : $e[n]$
- After precisely one DWA cycle, $s[n] = S(e[n]) = 0$.
- If $P = \sum |w_i - 1|$, then $|s[n]| < P/2$
- So $|e[n]|_{\text{avg}} = |s[n]| / n < P/(2n) \rightarrow 0$ as $n \rightarrow \infty$
i.e. the error tends to zero over several cycles
- **Assumption:** since $s[n]$ is bounded (like q-noise), we can assume that for "busy" signals it behaves like a bounded white noise
- Note that $e[n] = s[n] - s[n-1]$
- Then, $s[n]$ is white $\Rightarrow$ **$e[n]$ has 1st order noise-shaping**

Exercise 3.2 - Extra

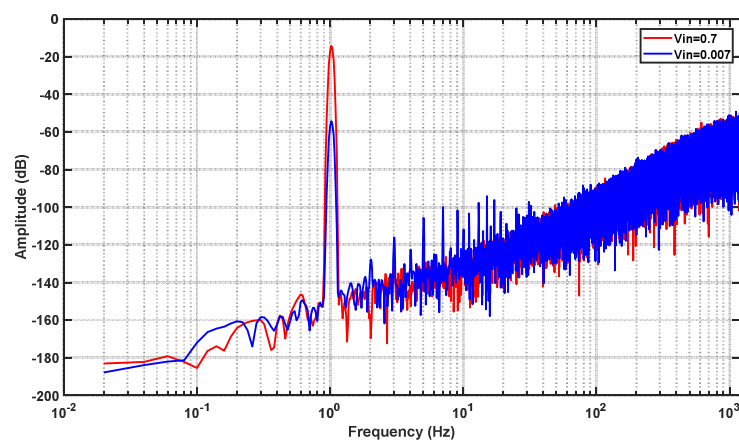
- Modify MOD2_3Bit_ThermoDAC.m to add DWA
 - Notice that you need to keep memory between different cycles.
- Observe how the error in the DAC appears now in the output spectrum.

Spectrum with DWA



- Spectrum with DWA (Data Weighted Averaging)
- No thermal noise

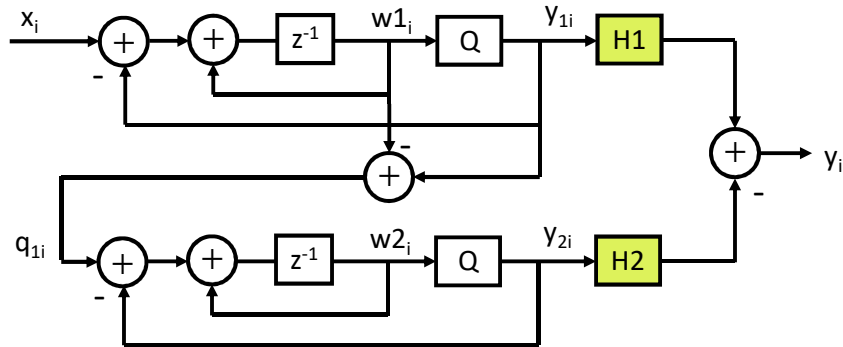
Spectrum with DWA



- For low input amplitudes DWA exhibits tonal behaviour

1-1 MASH (Multi-stage noise SHaping)

- Higher-order shaping with more lower-order loops



$$Y(z) = H_1 STF_1 X(z) + H_1 NTF_1 Q_1(z) - H_2 STF_2 Q_1(z) - H_2 NTF_2 Q_2(z)$$

Conditions:

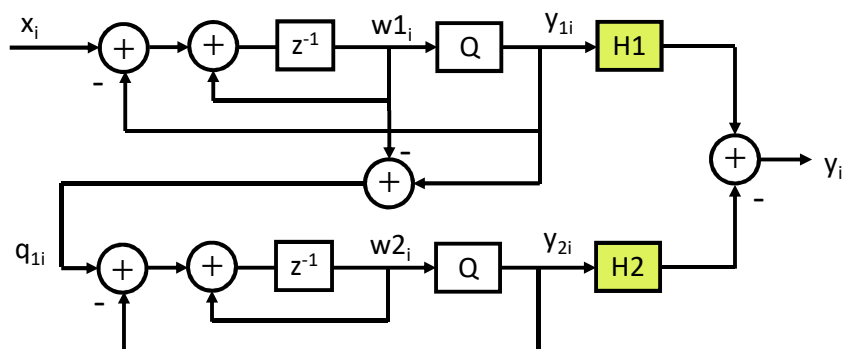
$$H_2 = NTF_1$$

$$H_1 = STF_1$$

Exercise 3.3

- Download (from Blackboard) the m-file: MOD1_for_MASH.m
 - This m-file simulates a 1st order modulator with a sinusoidal input.
 - Modify the file to implement a 1-1 MASH
 - Plot the output spectrum and show a 2nd-order noise shaping
 - Plot the output spectrum of 1st modulator and 2nd modulator. Which noise shaping do they show? Why?
- Add noise to the modulator and size it for SQNR=100 dB and SNR = 90 dB over a 20-kHz BW (as Assignment 3)
 - Optimize the input amplitude for maximum SNR
 - What is the difference with a single-loop 2nd-order modulator?
- Add leakage to the integrator by lowering the value of alpha
 - What do you observe? Why?

1-1 MASH



$$Y(z) = H_1 STF_1 X(z) + H_1 NTF_1 Q_1(z) - H_2 STF_2 Q_1(z) - H_2 NTF_2 Q_2(z)$$

Conditions: $H_2 = NTF_1$ In this case: $NTF_1 = NTF_2 = 1 - z^{-1}$
 $H_1 = STF_1$ $STF_1 = STF_2 = z^{-1}$

Exercise 3.3

• Matlab implementation

```
w1(i) = w1(i-1) + in(i-1) - y1(i-1);
y1(i) = 2*(w1(i)>=0)-1;
```

```
q1(i) = y1(i) - w1(i);
w2(i) = w2(i-1) + q1(i-1) - y2(i-1);
y2(i) = 2*(w2(i)>=0)-1;
```

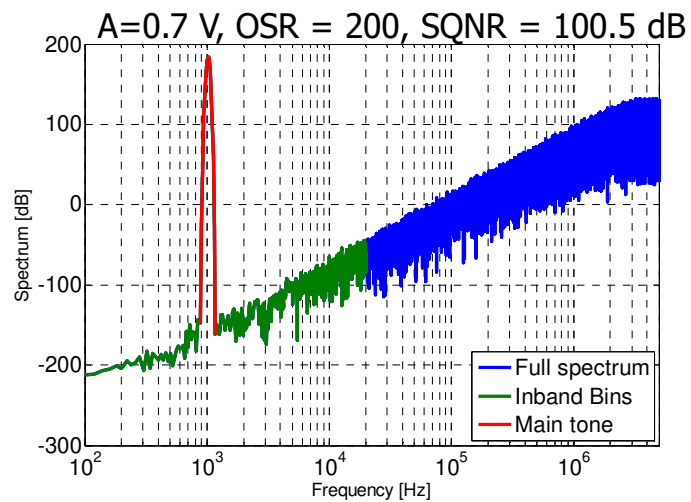
```
y(i) = y1(i-1)-(y2(i)-y2(i-1));
```

First stage

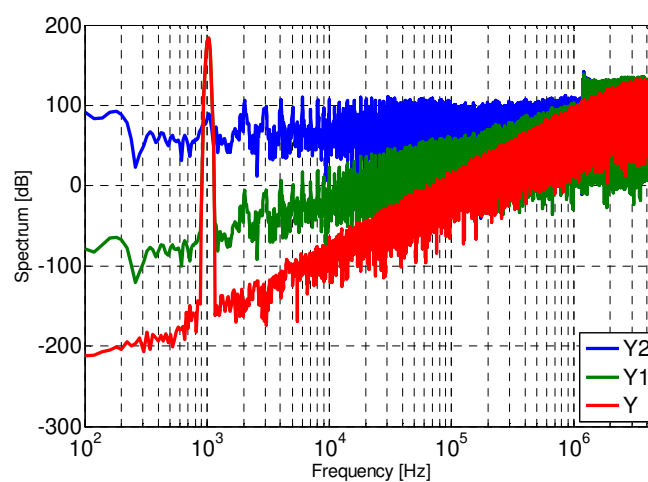
Second stage

Digital filtering

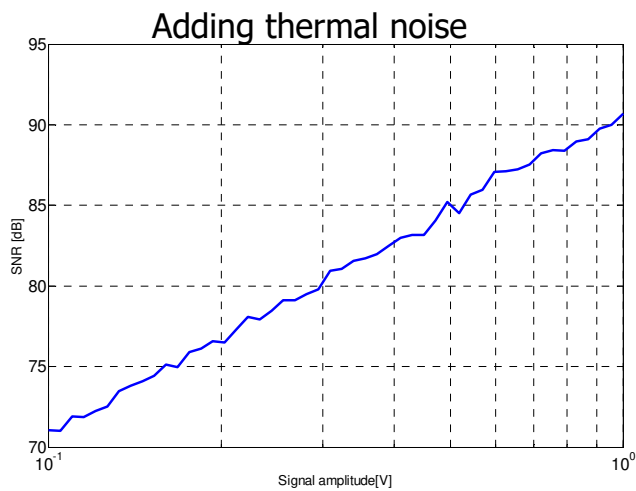
MASH – Output spectrum



MASH – Output spectra

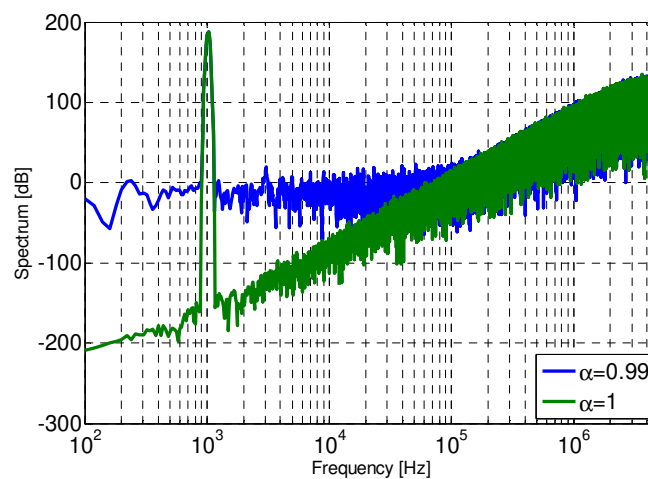


MASH – SNR



MASH – Integrator leakage

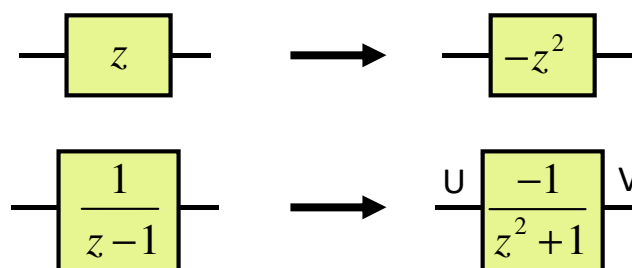
$A = 0.9 \text{ V}$



Today Overview

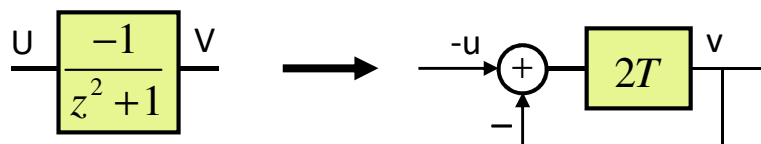
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Low-Pass to Band-Pass Transform (1)



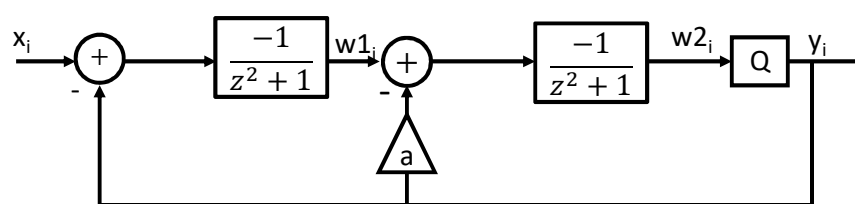
- Converts a pole at the origin i.e. $(z-1)$ to poles at $\pm\pi/2$ i.e. (z^2+1)

Low-Pass to Band-Pass Transform (2)



- Then, $(z^2 + 1)V(z) = -U(z) \Rightarrow V(z) = -z^{-2}V(z) - z^{-2}U(z)$
- Which results in the difference equation: $v_i = -(v_{i-2} + u_{i-2})$

Band-pass Modulator (1)



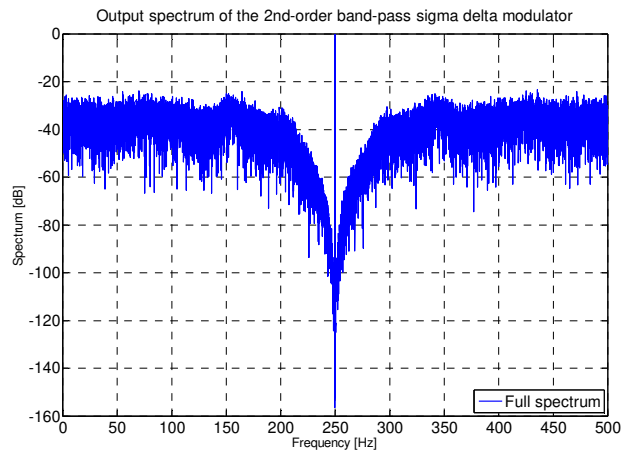
Difference equations are then:

$$w2_i = -w2_{i-2} - (w1_{i-2} - ay_{i-2})$$

$$w1_i = -w1_{i-2} - (x_{i-2} + y_{i-2})$$

$$y_i = Q(w2_i)$$

Band-Pass Modulator (2)



- The NTF's notch is at $f_s / 4$. Why?

Exercise 3.5

- Download (from Blackboard) and run the m-file: MOD2_BP.m
 - This file models a BP sigma delta modulator which was derived from the feedback modulator in Exercise 2.3 (the code for the low-pass modulator is kept but commented).
 - Modify the code so that it calculates the SNR into a certain bandwidth f_b which is symmetrically located around $f_s/4$.
 - What is the difference between the SNR of the low-pass modulator and the SNR of the band-pass modulator?